SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: STATISTICS WITH R PROGRAMMING **COURSE CODE: ITA0404**

COURSE OUTCOME COVERED

CO2 -- ASSESSMENT TOOL 2

CO2: Implement looping constructs, control structures, and recursive functions in R to solve computational problems with efficient program logic.Analyse the Correlation between the parameters in a dataset and establish the analysis (k3)

Assessment Tool Weightage: ***40%***

QUESTIONS: 5 **Total Marks: 50** **Duration: 1 Hour**

Coding Challenge

Code of Conduct:

I B.Vyshnavi Reddy (192525159) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Reg. No: 192525159

RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30						
Accuracy of Answer	25						
Application of Knowledge	20						
Completeness of Response	15						
Clarity & Presentation	10						

Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50

SAVEETHA SCHOOL OF ENGINEERING SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

1. Employee Bonus Calculator

Write an R function bonus() that accepts an employee's **salary** and **performance rating**.

- If rating ≥ 4, give a 10% bonus.
- Otherwise, give a 5% bonus.
- Use if-else.
- Return the bonus amount.
- Set the default rating as 3.

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2. Shopping Discount Calculator

Write an R function discount() that accepts the **purchase amount** and **membership status**.

- If the amount is greater than ₹5,000 **and** the customer is a member, give a 10% discount.
- Otherwise, give no discount.
- Use Boolean operators.
- Return the discount amount.
- Set the default membership status as FALSE.

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3. Loan Eligibility Checker

Write an R function loan_check() that accepts **age** and **income**.

A customer is eligible for a loan if:

- Age is 21 or above **and**
- Income is ₹30,000 or more. Use if-else and Boolean operators.

Return "Eligible" or "Not Eligible".

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4. Electricity Bill Calculator

Write an R function electricity_bill() that accepts the number of **units consumed**.

- Up to 100 units → ₹2 per unit.
- Above 100 units → ₹3 per unit.
- Add a default fixed charge of ₹100.
- Return the total bill.
- If units are negative, return "Invalid Units".

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5. ATM Withdrawal Checker

Write an R function withdraw() that accepts **balance** and **amount**.

- Check whether the amount is greater than 0.
- Check whether the amount is less than or equal to the balance.
- If both conditions are true, return the remaining balance.
- Otherwise, return "Transaction Failed".
- Set the default withdrawal amount as 1000.

Topics covered: if-else, arithmetic operators, Boolean operators, default arguments, and return values.

Bottom of Form

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 AT-2

NAME: B.Vyshnavi Reddy **FACULTY: Dr.G.Kumaresan**

REG.NO: 192525159 **COURSE CODE:ITA0404**

COURSE NAME: Statistics with R Programming

1.

Aim: To write an R function bonus() to calculate an employee's bonus based on salary and performance rating using if-else.

Code:

```
bonus <- function(salary, rating = 3)
{   if (rating >= 4) {     b <- salary *
0.10
  } else {
    b <- salary * 0.05
  }
return(b)
}

salary <- 50000
rating <- 4

print(bonus(salary, rating))
```

Input:

Salary = 50000
Rating = 4

Output:

[1] 5000

Result: The employee bonus is ₹5000.

2.

Aim: To write an R function discount() to calculate a 10% discount for members purchasing goods above ₹5000.

Code:

```
discount <- function(amount, member = FALSE) {
if (amount > 5000 && member == TRUE) {     d
<- amount * 0.10
  } else {
d <- 0   }
return(d)
}

amount <- 6000 member
<- TRUE

print(discount(amount, member))
```

Input:
Purchase Amount = 6000
Membership Status = TRUE

Output:
[1] 600

Result:
The discount amount is ₹600.

3.

Aim: To write an R function loan_check() to check loan eligibility based on age and income using Boolean operators.

Code:
```
loan_check <- function(age, income) {
if (age >= 21 && income >= 30000) {
return("Eligible")
  } else {     return("Not
Eligible")
  }
}

age <- 25 income
<- 40000

print(loan_check(age, income))
```

Input:
Age = 25
Income = 40000

Output:
[1] "Eligible"

Result:
The customer is eligible for the loan.

4.

Aim: To write an R function electricity_bill() to calculate the electricity bill based on units consumed with a fixed charge.

Code:
```
electricity_bill <- function(units, fixed = 100) {
if (units < 0) {     return("Invalid Units")   }
else if (units <= 100) {     bill <- units * 2 +
fixed
  } else {     bill <- units *
3 + fixed
  }
  return(bill)
}

units <- 150

print(electricity_bill(units))
```

Input:
Units Consumed = 150

Output:
[1] 550

Result:
The total electricity bill is ₹550.

5.

Aim: To write an R function withdraw() to check whether an ATM withdrawal is possible based on balance and withdrawal amount.

Code:
```
withdraw <- function(balance, amount = 1000) {
if (amount > 0 && amount <= balance) {
return(balance - amount)
```

```
  } else {
return("Transaction Failed")
  }
}

balance <- 5000 amount
<- 1000

print(withdraw(balance, amount))
```

Input:
Balance = 5000
Withdrawal Amount = 1000

Output:
[1] 4000

Result:
The withdrawal is successful and the remaining balance is ₹4000.